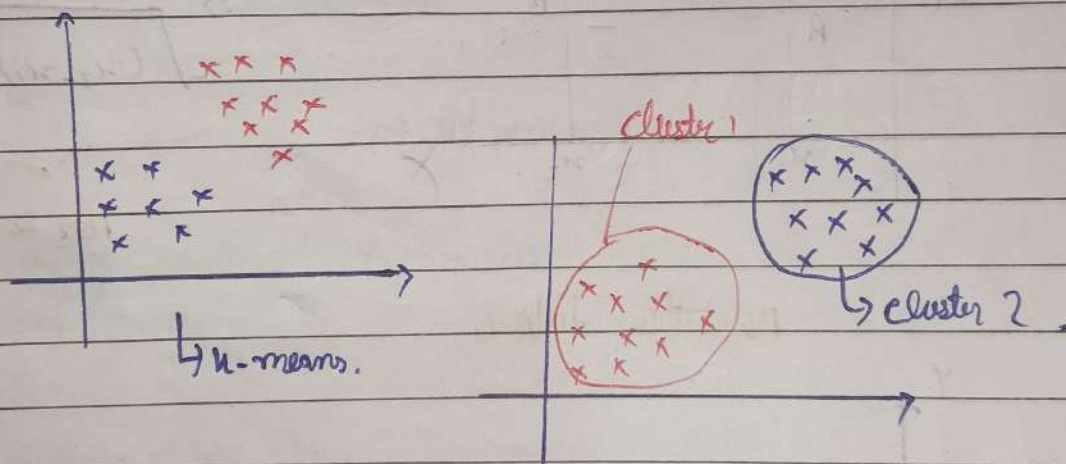


Unsupervised Learning (Data without label)

→ ① Clustering

- K-means
- K-median
- Hierarchical clustering
- Expectation maximization

Math Behind K-means Clustering

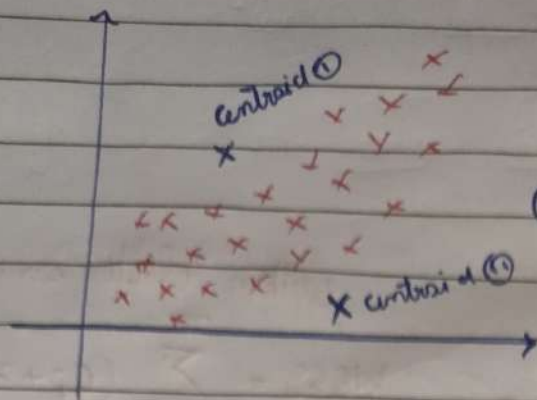


• Use // use

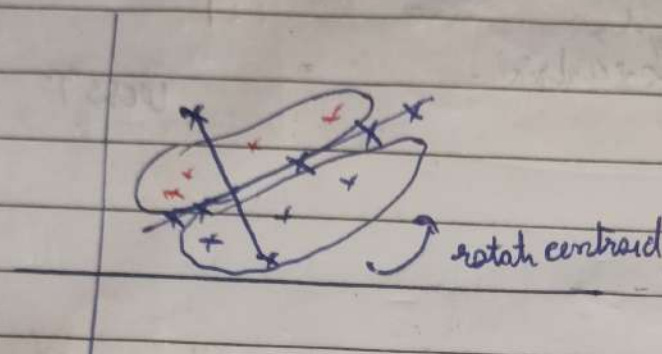
→ Algorithms

→ Metrics (Euclidean, Manhattan)

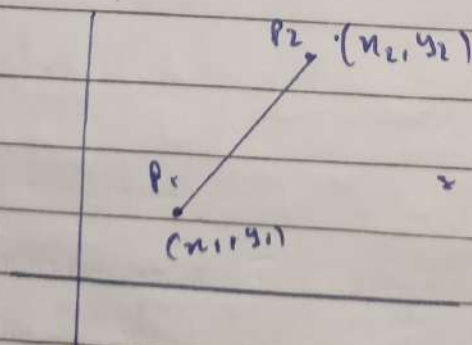
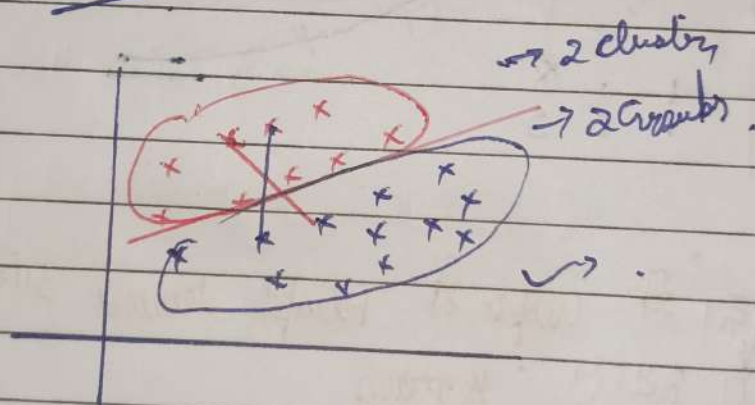
→ Elbow method → selecting k values.



- ① K-value $\Rightarrow$ Centroids $k=2$
- ② Initialize the centroid randomly
- ③ select grp & find mean values



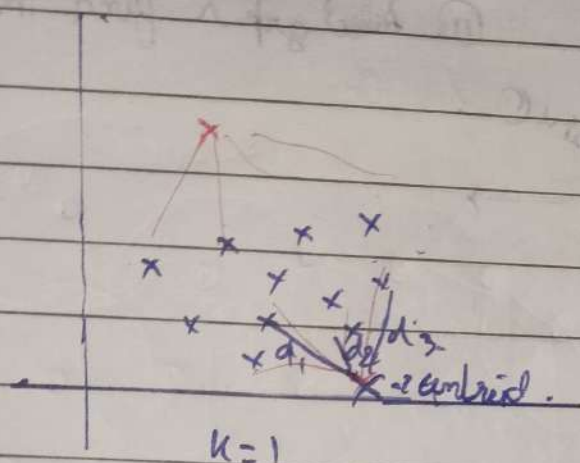
New Cent.



$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Elbow Method

$K=1$ to 20



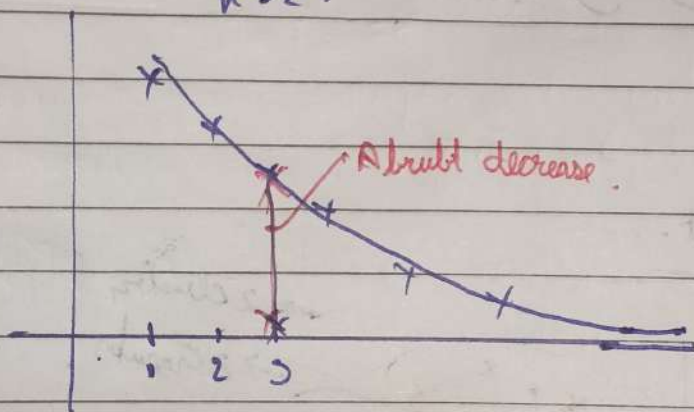
within cluster sum of squares

$$WCSS = \sum_{i=1}^n (C_i + x_i)^2$$

$K=1$

WCSS ↑

$K=2$



K means = 3

चूंकि ये Graph में Abrupt decrease आ रहा है.
तो हमारा K mean